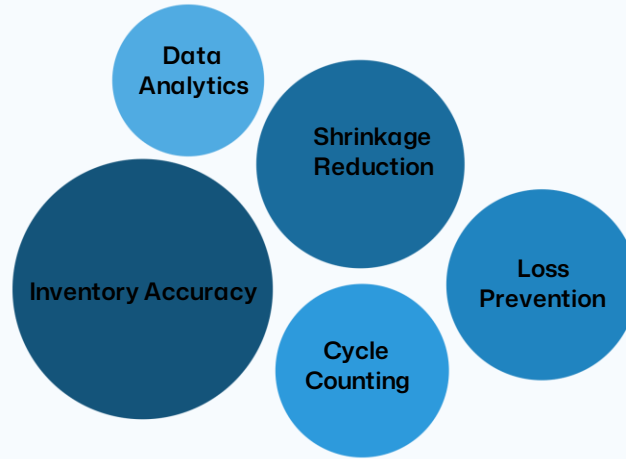


MetroFresh: Shrinkage & Inventory Accuracy Analysis



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Amdari Internship 2026
Power BI Capstone Project

Business Introduction & Overview

MetroFresh — the corner store, reimagined.

Across nine Midwestern states, 482 MetroFresh stores serve 1.8 million customers every week. Behind the counter sits a 4.2-million-member loyalty community and a hot-food program, FreshFast, that customers plan their day around. We don't just sell convenience — we sell *trust*: the right product, on the shelf, every time.

A high-volume, low-margin business where every unit counts.

Fuel, tobacco, drinks, snacks — and FreshFast, which alone drives 31% of in-store revenue. It's a model built on speed and turnover, which means one thing matters above all else: knowing exactly what's on the shelf. When inventory is right, the whole machine runs. When it's wrong, money leaks quietly from every aisle.

Business Problem

We were losing stock we couldn't see — and accepting it.

Shrink ran at 2.41% of sales, and inventory records were only 78% accurate — far below the 92% the business needs.

Worse, loss was treated as an unavoidable cost: counts were patchy, theft and vendor errors blurred into one

"unexplained" number, and the loss-prevention budget was spread by gut feel. We were managing the most theft-prone business in retail — on intuition.

Executive Summary

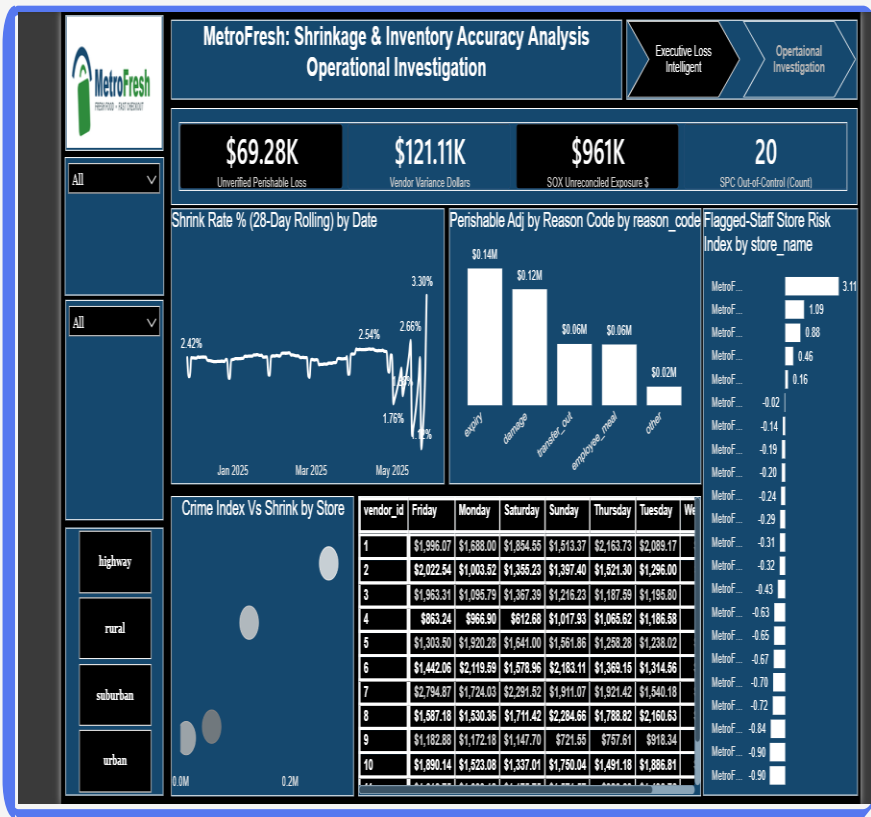
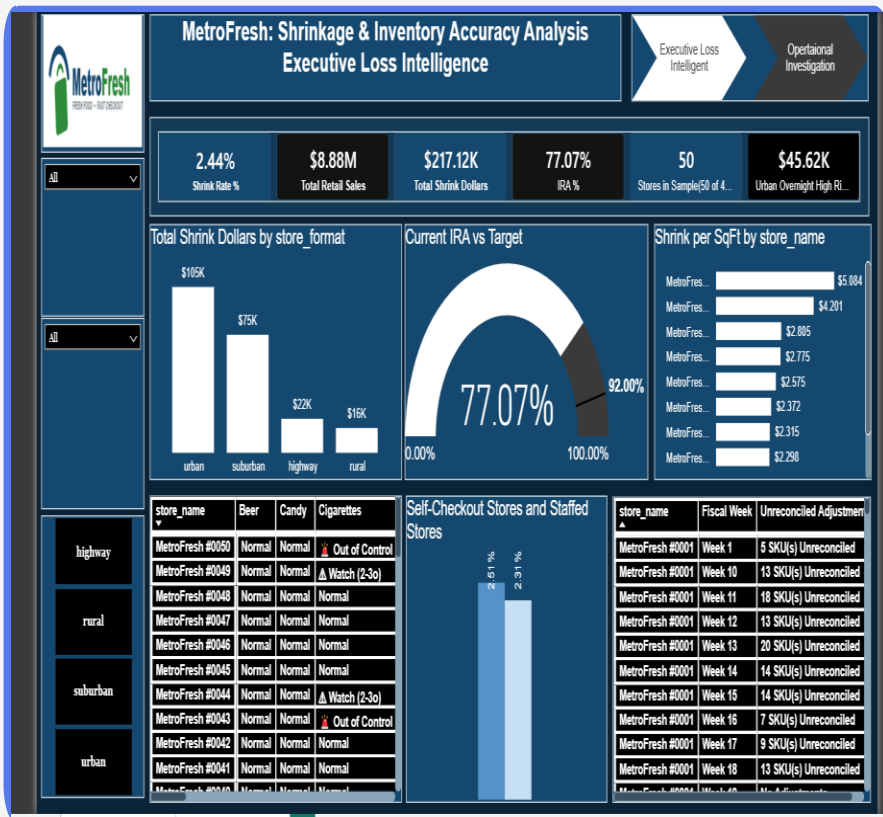
We turned shrink from a mystery into a map.

We built a single, reconciled source of truth and the picture sharpened instantly:

- Loss is concentrated, not everywhere — cigarettes and vape alone are 29% of all shrink, and they're our *least* accurate categories.
- The system now flags the exact hotspots: 20 store-categories statistically out of control (worst at 6.8σ) — a finite target list, not a chain-wide guess.
- The leaks have names: \$121K in vendor short-shipments, \$69K in perishable write-offs beyond what counts justify, and a \$961K unverified-adjustment gap now visible.
- Lifting accuracy from 78% to 92% isn't just recovery — it's the gate to auto-replenishment and online pickup, and the foundation of the brief's \$12–14M annual recovery opportunity.

We moved MetroFresh from "shrink happens" to "shrink is managed."

Executive Dashboards



Insights

Executive page

2.44% Shrink · \$8.88M Sales · \$217.12K Shrink \$ · 77.07% IRA · 50 of 482 stores · \$45.62K Urban Overnight)

Insight 1 — Real money is leaking through a blurry lens.

On \$8.88M of sales we lose \$217K (2.44%) — but with IRA at just 77.07%, we can't fully trust where that loss goes. The cards together tell the core story: *material dollars, low visibility*. Closing the 15-point gap to the 92% target is what turns the \$217K from "unexplained" into "actionable."

Insight 2 — One day-part already accounts for a fifth of the loss.

Urban Overnight High Risk = \$45.62K, which is ~21% of the entire \$217K shrink sitting in a single format-and-shift pocket. That's the clearest "start here" signal on the page — and the evidence to move loss-prevention budget toward urban overnight rather than spreading it evenly.

Operational page

(\$69.28K Unverified Perishable · \$121.11K Vendor Variance · \$961K SOX Exposure · 20 SPC Out-of-Control)

Insight 1 — Loss now has names and price tags.

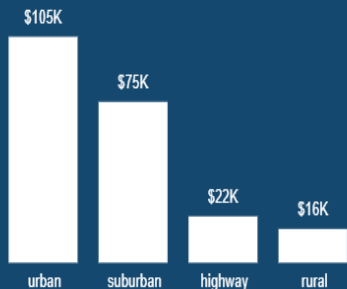
What used to be "unexplained" is now itemized: \$121.11K vendor short-shipments, \$69.28K perishable write-off beyond what counts justify, and 20 store-categories statistically out of control. The single largest *operational* leak is vendors (\$121K) — bigger than perishable — so receiving discipline is the highest-ROI fix on this page.

Insight 2 — The biggest exposure isn't theft, it's governance.

\$961K of unverified adjustments dwarfs every operational leak combined. The standout risk on this strip is audit defensibility, not shrinkage — a same-week reconciliation rule converts that \$961K blind spot into a SOX-defensible trail, and it's the cheapest of the fixes (a process rule, not new spend).

Insights

Total Shrink Dollars by store_format

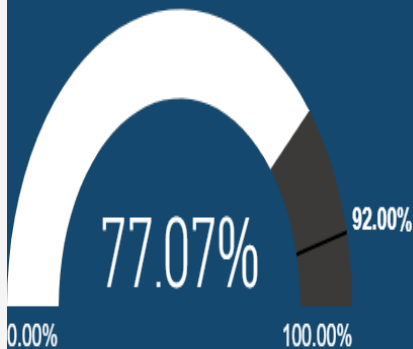


Total Shrink Dollars by store format → **Concentration of**

loss

1. Urban stores alone lose \$105K — more than suburban (\$75K), highway (\$22K) and rural (\$16K) combined. Urban + suburban = ~83% of all measured shrink
2. Highway + rural together are just \$38K — low-loss formats. The case to pull generic security spend off these and aim it at urban/suburban writes itself.

Current IRA vs Target



Current IRA vs Target (gauge) → **Strategic readiness**

/ IRA gap

1. At 77.07% vs the 92% target, accuracy sits ~15 points short of the gate for auto-replenishment and BOPIS — this is the single biggest blocker to modernization.
2. Roughly 1 in 4 counts is still off by more than a unit — each one a potential out-of-stock or hidden loss. Fixing accuracy is the foundation everything else stands on.

Shrink per SqFt by store_name



Shrink per SqFt by store_name → **Capital**

allocation

1. The worst store bleeds \$5.08 per square foot — over 2x the lowest visible store (\$2.30). Loss *density*, not store size, exposes the true problem locations.
2. Ranking by \$/sqft surfaces small, high-intensity stores a raw-dollar view would bury — a sharper target for the loss-prevention budget.

store_name	Beer	Candy	Cigarettes
MetroFres #0050	Normal	Normal	🔥 Out of Control
MetroFres #0049	Normal	Normal	⚠ Watch (2-3σ)
MetroFres #0048	Normal	Normal	Normal
MetroFres #0047	Normal	Normal	Normal
MetroFres #0046	Normal	Normal	Normal
MetroFres #0045	Normal	Normal	Normal
MetroFres #0044	Normal	Normal	⚠ Watch (2-3σ)
MetroFres #0043	Normal	Normal	🔥 Out of Control
MetroFres #0042	Normal	Normal	Normal
MetroFres #0041	Normal	Normal	Normal
MetroFres #0040	Normal	Normal	Normal

72hr Alert Status — Store × Category (SPC)

→ Anomaly detection

1. Cigarettes drives the red — stores #0050 and #0043 are "Out of Control (>3σ)," the worst at 6.8σ. The system names the exact hotspot to hit *within days*, not at the next count.
2. 20 store-category cells (of 750) are out of control — a finite, named target list that replaces guesswork and flips loss prevention from reactive to proactive.

Self-Checkout Stores and Staffed Stores



Self-Checkout vs Staffed Stores → *Named cause:*

technology

1. Self-checkout stores run 2.51% vs 2.31% staffed — a +0.20pp gap, confirming self-checkout adds loss.
2. But the gap is modest — so the answer is targeted camera/monitoring at SC lanes, not rolling back self-checkout. Evidence over assumption.

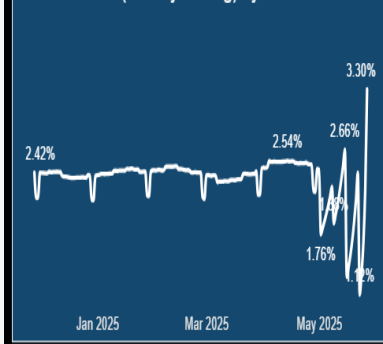
store_name	Fiscal Week	Unreconciled Adjustmen
MetroFresh #0001	Week 1	5 SKU(s) Unreconciled
MetroFresh #0001	Week 10	13 SKU(s) Unreconciled
MetroFresh #0001	Week 11	18 SKU(s) Unreconciled
MetroFresh #0001	Week 12	13 SKU(s) Unreconciled
MetroFresh #0001	Week 13	20 SKU(s) Unreconciled
MetroFresh #0001	Week 14	14 SKU(s) Unreconciled
MetroFresh #0001	Week 15	14 SKU(s) Unreconciled
MetroFresh #0001	Week 16	7 SKU(s) Unreconciled
MetroFresh #0001	Week 17	9 SKU(s) Unreconciled
MetroFresh #0001	Week 18	13 SKU(s) Unreconciled
MetroFresh #0001	Week 19	13 SKU(s) Unreconciled

SOX Gap — Stores Requiring Audit Action

→ *Governance / compliance*

1. Every listed store-week shows unverified adjustments (e.g., #0005 = 65 SKUs, #0003 = 53) — manual changes with no same-week count behind them.
2. A same-week reconciliation rule turns this \$961K blind spot into an auditable trail — exactly what a SOX-defensible posture requires.

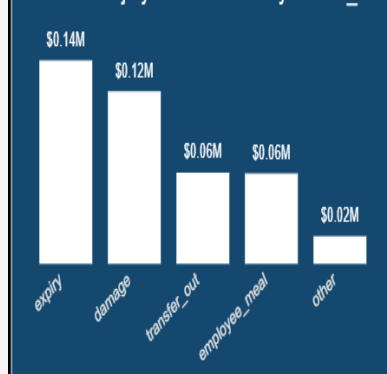
Shrink Rate % (28-Day Rolling) by Date



Shrink Rate % (28-Day Rolling) → *Sizing & monitoring the problem*

1. Shrink holds a steady ~2.4% band — this is *structural*, not random noise. It will not correct itself without intervention.
2. Because it's now near-real-time, a sustained rise above the band is an early warning — monitoring in days, not monthly hindsight.

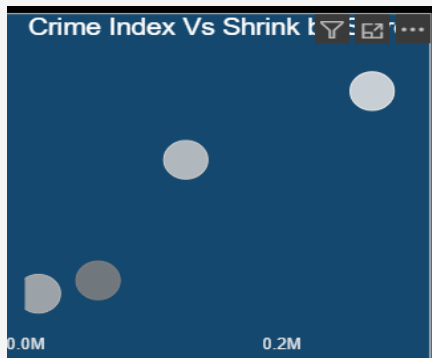
Perishable Adj by Reason Code by reason_code



Perishable Adj by Reason Code → *Named cause:*

perishable strain

1. Expiry (\$0.14M) and damage (\$0.12M) dominate write-offs — the FreshFast perishable strain in plain sight.
2. Yet only \$69.28K is "unverified" beyond what counts justify; damage/expiry are the easiest codes to hide theft behind — so a count-before-write-off rule separates true spoilage from masked loss.



Crime Index Vs Shrink by Store → External risk factor

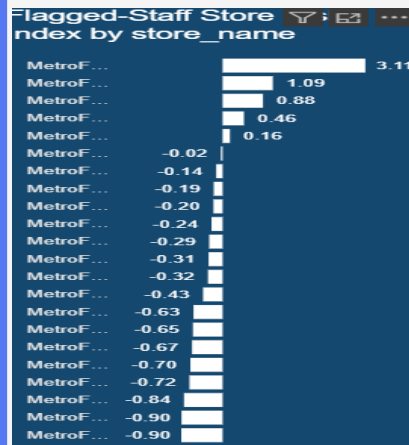
1. The higher-shrink bubbles sit toward the higher-crime side — external environment is a genuine risk factor, supporting targeted security where crime is highest.
2. But loss also shows up at lower-crime stores — crime alone doesn't explain it, so internal causes (vendor, perishable, scanning) must be tackled in parallel.

vendor_id	Friday	Monday	Saturday	Sunday	Thursday	Tuesday	Wednesday
1	\$1,996.07	\$1,688.00	\$1,854.65	\$1,513.37	\$2,163.73	\$2,089.17	\$1,854.65
2	\$2,022.54	\$1,003.52	\$1,355.23	\$1,397.40	\$1,521.30	\$1,296.00	\$1,355.23
3	\$1,963.31	\$1,095.79	\$1,367.39	\$1,216.23	\$1,187.59	\$1,195.80	\$1,367.39
4	\$863.24	\$966.90	\$612.68	\$1,017.93	\$1,065.62	\$1,186.58	\$612.68
5	\$1,303.50	\$1,920.28	\$1,641.00	\$1,561.86	\$1,258.28	\$1,238.02	\$1,641.00
6	\$1,442.06	\$2,119.59	\$1,578.96	\$2,183.11	\$1,369.15	\$1,314.56	\$1,578.96
7	\$2,794.87	\$1,724.03	\$2,291.52	\$1,911.07	\$1,921.42	\$1,540.18	\$2,291.52
8	\$1,587.18	\$1,530.36	\$1,711.42	\$2,284.66	\$1,788.82	\$2,160.63	\$1,711.42
9	\$1,182.88	\$1,172.18	\$1,147.70	\$721.55	\$757.61	\$918.34	\$1,147.70
10	\$1,890.14	\$1,523.08	\$1,337.01	\$1,750.04	\$1,491.18	\$1,886.81	\$1,337.01

Vendor Short-Shipment matrix → Named cause:

vendor accountability

1. Vendor 7 is the standout — \$2,794.87 short on Fridays (the largest single cell) and the worst vendor overall (\$14.3K, 5.0% short rate). Friday deliveries need a blind receiving check.
2. Variance is spread across all vendors (~4.7–5.1% short rates), totaling \$121K — this is a *receiving-process* gap, not one rogue supplier. Fix the SOP, not just the vendor.



Flagged-Staff Store Risk Index → Named cause:

behavioral (governance-safe)

1. One store stands out at +3.1σ on negative-adjustment intensity — a statistical outlier worth an HR/Legal-gated review (aggregate, never individual).
2. Most stores cluster near or below zero — the signal is concentrated, not widespread, so intervention stays targeted and defensible, without blanket suspicion.

Actionable Recommendation

Five moves — all within budget, all startable now.

1.Count what gets stolen. Weekly cycle counts on tobacco, vape and beer — lift their accuracy from the high-50s toward 90%.

Labor-neutral.

2.Attack the 20, not the 482. Send loss-prevention straight to the out-of-control store-categories — surgical spend on the ~29% of loss that hurts most.

3.Make vendors prove the count. A blind receiving check against the PO on the top suppliers — recover the \$121K leak.

4.Verify before you write off. A count before any high-value perishable write-off — separate true spoilage from theft hiding inside it.

5.Close the audit gap. A same-week reconciliation rule on adjusted items — retire the \$961K exposure with an auditable trail.

Same budget. Same team. A clear target list instead of a guess.

Next Step: A 6-Week Vendor-Receiving Pilot

Don't roll a new process out to 482 stores and hope. Test the biggest leak on a few stores, measure against a control, and let the results decide.

The setup — Week 0

- Pick the stores that receive from Vendor 7 (our biggest named leak: \$121K, 5% short rate)
- Pair them by type and size; in each pair, one store counts deliveries against the order, its twin changes nothing
- Lock today's numbers first: short-ship rate, dollars, shrink rate

Run it — Weeks 1 to 6

- Counting stores check every delivery against the PO and short-pay the gaps
- Extra attention on Fridays (where the worst variance showed up)
- Watch weekly: short-ship rate vs the control, and receiving speed

Call it a win if — by Week 6

- The counting stores beat the control
- The gap holds across most of the six weeks, not one lucky week
- Receiving didn't slow down

The decision — Week 7

- Worked → scale to the top 5 vendors, then the chain
- Half-worked → tighten the process and extend
- No difference → move to the next leak

Keep it honest

- The 50-store sample proves the method; the recovery number is confirmed at full-chain scale
- Vendor variance is a cost figure — keep it separate from the \$217K cycle-count shrink